Captain’s Summer Exchange - Tesla STEM High School - Repaired practice edition - Answer key

Division C | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

27. (6 points) Authored replacement for source item Question 27: Solve this case and explain your reasoning: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

reasoning: Gypsum is Mohs 2; fluorite is Mohs 4.
The test brackets hardness between about 2 and 4.
Calcite at 3 is compatible, but additional evidence is needed. (3 points).

conclusion: A bracket is evidence, not a unique identification. (3 points).

60. (6 points) Authored replacement for source item Question 60: Solve this case and explain your reasoning: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

reasoning: Compare hematite and magnetite rather than using color.
Hematite commonly gives reddish-brown streak; magnetite is strongly magnetic.
Favor hematite while checking other properties. (3 points).

conclusion: Use diagnostic combinations. (3 points).

70. (4 points) Authored replacement for source item Question 70: Differentiate mineral and rock cleavage.

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

answer: Mineral cleavage is crystal breakage; rock cleavage is an aggregate planar fabric. (4 points).

95. (6 points) Authored replacement for source item Question 95: Solve this case and explain your reasoning: A massive specimen lacks visible faces. Does it lack crystalline order?

reasoning: Habit describes external appearance.
Many small intergrown crystals can produce a massive appearance.
Other structural and physical observations are needed. (3 points).

conclusion: Absent faces do not establish an amorphous material. (3 points).

96. (6 points) Authored replacement for source item Question 96: Solve this case and explain your reasoning: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

reasoning: Each tetrahedron has four oxygen corners.
Each corner is shared by two tetrahedra.
Count four halves = two oxygen atoms per silicon. (3 points).

conclusion: Sharing changes atom bookkeeping. (3 points).

123. (4 points) Authored replacement for source item Question 123: What is a pseudomorph?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

answer: A replacement mineral preserves the external form of an earlier material. (4 points).

124. (4 points) Authored replacement for source item Question 124: What do vesicles record?

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

answer: Gas bubbles trapped during solidification, without uniquely fixing composition. (4 points).